

List of C Programs

A. Operators, data types.

1. Write a C Program to display "Hello World" on screen.
2. Write a C Program to display your name and address in different lines.
3. Write a C program to calculate sum of two integers numbers and print the result.
4. Write a C Program to display square and cube of a given number.
5. Write a C Program to find area of circle. ($A = \pi r^2$)
6. Write a C Program to find area of a triangle (Hint : $A = \frac{1}{2} * h * b$).
7. Write a C Program to compute simple interest ($SI = PTR/100$).
8. Write a C Program to interchange values of two variables using third variable.
9. Write a C Program to interchange values of two variables without using third variable.
10. Write a C Program to convert Km value into meter value.
11. Write a C program to convert seconds into hours, minutes and seconds and vice-versa.
Ex: i/p 5000sec o/p: 1 Hour , 23mins , 20secs .
12. Write a C Program to input marks of three subjects and display total and percentage.
13. Write a C Program to input quantity and price of an item and display total Rs. on screen. Also input discount (%) and display final payment Rs. on screen.
14. Write a C Program to accept any character from user and display its ASCII number on screen.
15. Write a C Program to input any 2 digit ASCII number and display appropriate character on screen.
16. Write a C Program to convert Celsius value into Fahrenheit value. (Formula : $F = C * 9/5 + 32$)
17. Write a C Program to convert Fahrenheit value into Celsius value. (Formula : $C = (F - 32) * 5/9$)
18. Write a C Program to input two digit number from user and display reverse of that number on screen
(Hint : I/P : 32 O/P : 23)
19. Write a C Program to input three digit number from user and calculate sum of first and last digit.
(Hint : I/p : 358 O/p : 11)
20. Write a C program to input a 4 digit number and display the ascii value of each digit.
(I/P : 1234 O/P: 49 50 51 52)

B. Conditional statements

1. Write a C Program to check whether the entered number is odd or even.
2. Write a C Program to display greatest of three numbers.
3. Write a C Program to read any character from user and if that character is capital letter then display it with small letter and if it is small letter then display with Capital letter.
4. WAP to input any char and display whether it is capital letter or small letter or digit or special character .
5. Write a C Program to accept marks of 3 subjects and display total marks, percentage and class depending upon following conditions. Less than 35% fail , 35% - 49% pass , 50% - 59% 2nd class , 60% - 69% 1st class, 70% & above FCD.
5. Write a C Program to check whether the entered year is leap year or not.
6. Write a C Program to input price, quantity of an item from user and display final payment according to following condition. If total purchase is greater than 1000Rs. then calculate discount 15% otherwise calculate 10% of total rs.
7. Write a C Program to simulate a simple calculator using switch statement. The program should read two numbers from user and an operator and then perform respective arithmetic operation and print the final result.
8. Write a C program to check whether a given number is Armstrong number or not. (hint : $153=1^3 + 5^3 + 3^3$)
9. Write a C program to display three numbers in ascending order using if-else statement.
10. Write a program to read three sides of a triangle and check whether they form a triangle or not
11. WAP to find the roots of a given quadratic equation ($ax^2 + bx^2 + c = 0$) by reading the coefficients a, b, c.
12. Calculate electric bill depending on the following conditions

Consumption Units	Rate of Charge
000 - 200	Rs 0.50 per unit
201 - 400	Rs 100 plus Rs.0.65 per unit excess of 200
401 - 600	Rs. 230 plus Rs.0.80 per unit excess of 400
601 and above	Rs. 390 plus Rs.1.00 per unit excess of 600

C. Control statements

1. WAP to Print first N numbers.
2. Wap to calculate sum of first N numbers.
3. WAP to print first N even or odd numbers.
4. WAP to print even or odd numbers within a given range.
5. WAP to find factorial of a given number.
6. WAP to print first N fibonacci numbers.
7. WAP to find gcd & lcm of two numbers.
8. WAP to check whether a given number is prime or not.
9. WAP to print first N prime numbers.
10. Wap to find the number of binary ones in a given number.(eg:182 no of ones=5).
11. WAP to check whether a given number is palindrome or not.
12. WAP to check whether a given number is bitwise palindrome or not.
13. WAP to add all the digits of a given number(ex:12345 sum=15).
14. WAP to check whether a given number is power of 2 or not.
15. WAP to convert 2 digit number to word. Eg : 45 o/p :forty five.
16. Write a C program to print all Armstrong numbers between 1 to 500. (hint $153 = 1^3 + 5^3 + 3^3$).
17. Write a C program to print all Combinations of characters A, B, C.
18. Write a C program to find the factors of a given number.(eg:216 factors=2,3,4,6,8,9,12,18,24,27,36,54,72,108)
19. Write a C program to find the prime factors of a given number.(eg:216 prime factors=2,3)
20. Write a program to Convert a given decimal number to roman equivalent upto 2000.
(Hint: 1=I 4=IV 5=V 9=IX 10=X 40=XL 50=L 90=XC 100=C 400=CD 500=D 900=CM 1000=M)
21. write a program to add even digits of a given number.(eg : 13462 output: 4+6+2=12).
22. WAP to find gcd and lcm of 3 numbers.
23. WAP to calculate nCr and nPr . ($nCr = n! / (r! * (n-r)!)$ $nPr = n! / (n-r)!$)
24. WAP to find the product of two 2digit numbers by shift and add method.
25. WAP to read two numbers (n1,n2) and divide the first number(n1) by second number(n2) by shift and add method and display quotient and remainder.

Generate the following patterns:

1
22
333
4444
55555

b. 1
12
123
1234
12345

c.
A
B B
C C
D D
E E

d.
A
B A B
C B A B C
D C B A B C D
E D C B A B C D E

e.
1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1

f.

D. ARRAYS

1. *wap to read and print N numbers.*
2. *WAP to calculate average of N numbers.*
3. *WAP to add the two arrays and print the resultant array along with the input arrays.*
4. *WAP to count the no of Even and Odd numbers.*
5. *WAP to print the largest and smallest elements of an array.*
6. *WAP to count the occurrence of the given number.*
7. *WAP to reverse the given array and print both the arrays.*
8. *WAP to arrange the given numbers in Ascending/Descending order.*
9. *WAP to arrange the given numbers in Ascending order and perform binary search to search a key element .*
10. *WAP to find binary equivalent of the number. (eg: Num = 125 binary equivalent : 1111101).*
11. *WAP to find decimal equivalent of given binary number. (Eg: binary=1001011 decimal equivalent=75).*
12. *WAP to convert a given decimal number to hexadecimal and vice-versa.*
13. *Write a C program to delete an element from the 'n'th position of an array.*

Eg: given array 10 20 30 40 50 pos=3 resultant array 10 20 40 50

14. *Write a C program to Insert an element at the 'n' th position.*

Eg: given array 10 20 30 40 50 pos=3 element =66 resultant array 10 20 66 30 40 50

15. *WAP to delete a given number from the array.*

Eg: i/p array 1 2 3 2 5 2 10 2 6 , number to be deleted is 2 resultant array: 1 3 5 10 6.

16. *Write a C program to read a sequence of positive integers and store it in an array, then read an integer N, and then rotate clockwise the content of the array a corresponding number of positions and print resultant array. For example, if the sequence is 10,20,30,40,50 and the number 3 is entered, the array becomes 30,40,50,10,20*

17. *WAP a program to rotate clockwise only the odd numbers of an array by N number of positions.*

Eg: 1 2 3 5 7 2 2 9 2 13 N=3 o/p: 7 2 9 13 1 2 2 3 2 5

18. *write a program to calculate the lcm of n numbers.*

19. *WAP to read two arrays of size M&N and find the intersection of the these arrays and print the resultant array along with the input arrays.*

20. *WAP to read two arrays of size M&N and find the union of the these arrays and print the resultant array along with the input arrays.*

Two dimensional arrays

21. WAP to read a two-dimensional array of size $m \times n$ and print it in m rows and n columns.

22. WAP to read two $m \times n$ matrices and find the sum of the matrices and display the resultant matrix along with the input matrices.

22. WAP to find the determinant of the $m \times n$ matrix.

23. WAP to read two $m \times n$ matrices and find the product of the matrices and display the resultant matrix along with the input matrices.

E. STRINGS

1.WAP to read and print your name using gets() and puts() functions.

2.Implement all the string instructions.

a>strlen b>strcpy , strncpy c>strcmp , strncmp d>strcat , strncat.

3.WAP to input two strings and check whether the first string is equal or greater or lesser than second string .

4.WAP to count the number of vowels in a given string.

5.WAP to check whether a given string is palindrome or not.

6. WAP to read a string and print the count of all the characters.

Example : good morning output : g=2 o=3 d=1 m=1 r=1 n=2 i=1.

7. WAP to check whether a given sub string is present in the main string.

8.WAP to count the number of occurrence of given sub string is in the main string.

main string: ram is good good very good boy sub string :good o/p: count=3

9. write a program that accepts name and prints its short form.

example : Mahendra Singh Dhoni output : M S Dhoni

10. WAP to read a string and print all the words in reverse order

example: good morning india output : india morning good

11. WAP to delete multiple spaces in a given string.

Eg: good morning india op: good morning india.

12.WAP to read a password from the user and compare it with predefined password and display appropriate messages.

Two dimensional Strings

13.WAP to find the greatest of N strings.

14.WAP to read N names and arrange them in ascending order.

15.WAP to read N names and delete a given name from the N names.

16.WAP to read N names and arrange them in ascending order and then insert a given name in the appropriate position.

F. POINTERS

- 1.WAP to read a number using pointers and print its value along with its address.
- 2.WAP to add , sub, multiply and divide two numbers using pointers.
3. Write a C Program to swap the contents of two variables using pointers.
- 4.WAP to check whether a given number is even or odd.
5. WAP to input any char and display whether it is capital letter or small letter or digit or special character using pointer .
- 6.WAP to find the greatest of three numbers using pointers.
- 7.WAP to calculate the factorial of a number using pointers.
- 8.WAP to calculate gcd and lcm of two numbers using pointers.
9. WAP to check whether a given number is prime or not using pointers.
- 10.WAP to reverse a given number and print both reverse and the given number using pointer.

Arrays and Pointers

11. WAP to read and print N numbers using pointers.
12. WAP to add the two arrays and print the resultant array along with the input arrays using pointers.
13. WAP to reverse the given array and print both the arrays using pointers.
14. WAP to arrange the given numbers in Ascending/Descending order.
15. Write a C program to delete an element from the 'n'th position of an array.

Eg: given array 10 20 30 40 50 pos=3 resultant array 10 20 40 50

Strings and Pointers

- 16.WAP to find the string length using pointers.
- 17.WAP to count the occurrence of a given char in the string using pointers.
- 18.WAP to delete all the white spaces in a given string using pointers.
- 19.WAP to convert all the lower case letters to upper case and all the upper case letters to lower case and print both the strings using pointers.

G. FUNCTIONS

call by value

1. WAP to add two numbers, the two values are passed to the function which returns the sum.
2. WAP to find the greatest of three numbers, here the three values are passed to the function which returns the greatest.
3. WAP to find the factorial of a given number, here 'n' is passed to the function which returns the factorial of n.
4. WAP to calculate gcd of two numbers, here the two values are passed to the function which returns the gcd.
5. WAP to calculate lcm of three numbers.
6. WAP to reverse a given number and print both the numbers, here a number is passed to the function which returns the reverse of the number.
7. WAP to find the number of binary ones in a given number, here a number is passed to the function which returns the result(number of ones). (eg:182 no of ones=5).
8. WAP to implement 'pow' function, here the two values(x,n) are passed to the function which returns the nth power of the first number(x^n).
9. Write a C program to input a 4 digit number and display the ascii value of each digit, here a 4digit number is passed to the function which prints ascii value of each digit from left to right. (I/P :1234 O/P: 49 50 51 52).
10. WAP to calculate sine of a given number . here values (x in degrees) is passed to the function which returns sine of the number. (hint : $\sin(x) = x^1/1! - x^3/3! + x^5/5! - x^7/7! \dots$)

call by reference

11. WAP to read a number and pass its address to the function which prints its value along with its address.
12. WAP to read two numbers and pass their addresses to the function which computes the sum and difference and does not return anything , but still the sum and diff should be displayed in the main program.
(Hint: pass the addresses of all the four variables n1,n2,sum,diff).
13. WAP to find the greatest of three numbers, here the addresses of all the four variables(a,b,c,greatest) are passed to the function and the function will not return anything , but still the result should be displayed in the main program.
14. WAP to read two numbers and pass their addresses to the function which computes the gcd and lcm and does not return anything , but still the gcd and lcm should be displayed in the main program.
(Hint: pass the addresses of all the four variables n1,n2,gcd,lcm).
15. WAP to find the factorial of a given number, here the addresses of 'n' is passed to the function which returns the factorial of n.
16. WAP to check whether a given number is palindrome or not, here the addresses of number is passed to the function which returns the reverse of that number, and the given number is compared with the reverse to check for palindrome.

Arrays and Functions

17. WAP to calculate average of N numbers, here the size and the starting address of the array is passed to the function which returns the average.
18. WAP to reverse the given array and print both the arrays, here the address of both the arrays(a,rev), size are passed to the function which will copy the reverse of first array(a) into second array(rev).
19. Write a C program to delete an element from the 'n'th position of an array, here the address of the array,size and the Nth position are passed to the function which will delete the element from the nth position and resultant array is printed in the main program. Eg: given array 10 20 30 40 50 pos=3 resultant array 10 20 40 50.
20. WAP to find both largest and smallest elements of an array,here the function will not return anything but both largest and are to be printed in main program.
21. WAP to count the no of Even and Odd numbers, here the function will not return anything but both the counts are to be printed in main.

Strings and Functions

22. Implement all the string instructions using user defined functions.

a>strlen b>strcpy , strncpy c>strcmp , strncmp d>strcat , strncat.

23. WAP to count the occurrence of a given key element in the string, here starting address and key element are passed to the function which returns the count.